

Supplementary material

Active Contact Localization Without Joint-Torque or Force/Torque Sensing:
Ambiguity-Preserving Inference for Robot Manipulators

S1. Recursive versus batch estimation on identical windows

The profile likelihood of the main text (Eq. 9) can also be maximized without recursion. The batch estimator maximizes it over all windows collected so far by a global grid search over the search region $\mathcal{B}$ (20,000 points) followed by multi-start Nelder–Mead refinement. It reports a Gaussian covariance from the analytic profile Fisher information at the estimate, in which directions without information are capped at the size of $\mathcal{B}$, so that its uncertainty is rank-aware: after a single view, its reported radius spans the search region along the line of action.

To compare the two estimators on identical data, 400 additional trials (test seeds 50000–50399, disjoint from the main study) took five views without early stopping. The next view was chosen at random, so that the view sequence is independent of either estimator, or by the active rule of the CPF. Both estimators were evaluated after every view on the same measurement windows (Table S1).

Table S1: CPF and batch estimator on identical windows (random next views, 400 trials). Median and 95th-percentile position error, and coverage of the 95% credible ellipsoid.

Views	CPF (proposed)			Batch		
	Median [mm]	p95 [mm]	Coverage [%]	Median [mm]	p95 [mm]	Coverage [%]
1	51.0	148.7	86.2	100.1	224.3	99.8
2	11.6	35.3	87.8	8.4	45.9	99.2
3	9.6	30.7	87.2	7.2	25.2	97.5
4	7.9	24.3	88.8	6.3	19.9	95.2
5	7.4	21.0	88.5	5.9	18.5	91.5

The final accuracy was similar: after five views, the p95 error was 21.0 mm for the CPF and 18.5 mm for the batch estimator (paired bootstrap 95% CI of the difference: -0.2–5.0 mm). The batch estimator’s rank-aware covariance was better calibrated at every view.

A second comparison asks how often each estimator delivers an accurate report when the risk of a false report is controlled. The reporting rule reports at the first view $k \geq 2$ with $sr_{95} < 10$ mm. For each estimator, the scale s is calibrated on the even seeds as the value that maximizes accurate reports ($e_p \leq 10$ mm) subject to a per-trial false-report rate ($e_p > 10$ mm at the report, divided by the number of trials) whose 95% Clopper–Pearson upper bound is at most 5%, and it is then evaluated on the odd seeds (200 trials; Table S2). The constraint applies to the rate per trial, whose denominator includes the trials without a report; Table S2 gives the false reports both per trial and per report.

With random views, the batch estimator delivered an accurate report in 29.0% of the evaluation trials and the CPF in 6.5% (paired difference -22.5 percentage points, 95% CI -28.5–17.0). The particle posterior is optimistic in the tail (main text, Fig. 4), and a rule that must keep the false-report risk below 5% has to discount its radius heavily.

Table S2: Calibrated reporting on the evaluation half (200 trials per view-selection rule). Reports: number of trials that ended with a report. Accurate, false per trial, and none (no report): percentage of all evaluation trials; s : calibrated scale. False per report: percentage of the reports.

Next view	Estimator	s	Reports	Accurate [%]	False [%]		None [%]	Views
					per trial	per report		
Random	CPF	1.59	??	6.5	1.0	??	92.5	3.27
Random	Batch	0.75	??	29.0	1.0	??	70.0	3.98
Active (CPF)	CPF	1.59	??	16.0	1.5	??	82.5	3.29
Active (CPF)	Batch	1.06	??	44.0	1.0	??	55.0	3.47

Both estimators evaluate the same likelihood on the same windows, so their similar final accuracy confirms that the accuracy is set by the measurement model: the uncertainty budget, the marginalized follower variation, and the actively selected configurations. The recursive filter provides the posterior between views on which recovery and view selection act; the batch re-estimate provides the better-calibrated uncertainty for the reporting decision and the final result of an episode.